Yukta Kasina

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EDUCATION

Northeastern University, Boston, MA *Sep 2024 – May 2026*
Master of Science in Artificial Intelligence (GPA: 3.6)
Coursework: Machine Learning, NLP, Foundations of AI, Data Mining, Algorithms

SRM Institute of Science and Technology, India *Sep 2020 – Jun 2024*
Bachelor of Technology in Computer Science and Engineering (GPA: 3.74)
Coursework: AI, Data Structures & Algorithms, Computer Networks, DBMS, Data Mining & Analytics

SKILLS & TOOLS

Languages:	Python, C++, Java, SQL, TypeScript
ML/DL:	LLM Fine-tuning & Pretraining, Recommendation Systems, NLP, Deep Learning, Zero-shot Learning, Transformer Architectures
Frameworks:	PyTorch, TensorFlow, HuggingFace Transformers, LangChain, Scikit-learn
Data:	Data Gathering & Preprocessing, Pandas, NumPy, Matplotlib, Seaborn, Apache Spark
Cloud:	AWS (SageMaker, Lambda, S3), GCP
Other:	Model Evaluation Pipelines, Experimentation & Analysis, Research Documentation
Certifications:	Stanford ML Specialization, AWS ML Foundations, Oracle DB Foundations

PROFESSIONAL EXPERIENCE

Kirusa, Inc. *Aug 2023 – Nov 2023*
Machine Learning Intern
- Fine-tuned and deployed NLP models using HuggingFace Transformers, integrating zero-shot classification for text summarization tasks.
- Designed preprocessing pipelines for large-scale unstructured datasets, ensuring quality for downstream ML tasks.
- Collaborated with cross-functional teams to align AI solutions with business goals, improving accuracy by 12%.

AICTE *May 2023 – Jul 2023*
Machine Learning Intern
- Built deep learning-based recommendation components for image classification workflows.
- Developed model evaluation frameworks to assess performance against domain-specific benchmarks.
- Produced visual and written reports for both technical and non-technical stakeholders.

AICTE *Sep 2022 – Nov 2022*
Data Engineering Intern
- Developed automated ETL pipelines for large, heterogeneous datasets, improving analytics readiness by 20%.
- Integrated preprocessing steps including feature extraction, normalization, and data validation.

PROJECTS

Earthquake First Aftershock Prediction — *Random Forest, XGBoost, DNN, TensorFlow*
- Trained regression models on USGS seismic data (2004–2018) to predict aftershock timing and magnitude.
- Engineered temporal and spatial features to boost model performance across diverse earthquake zones.
- Validated models using MSE, MAE, and R^2, achieving improved accuracy over baselines.

Explainable AI for Tumor Classification — *ResNet, PyTorch, Grad-CAM, SHAP*
- Built ResNet/VGG models on preprocessed tumor image datasets to classify benign vs. malignant growths.
- Applied Grad-CAM and SHAP to visualize learned features, aiding model transparency and trust.
- Enhanced model generalization using data augmentation and 5-fold cross-validation techniques.

Reinforcement Learning for Pokémon Battles — *DQN, PPO, PyTorch, OpenAI Gym*
- Developed and trained DQN and PPO agents using reward signals to learn optimal game strategies.
- Implemented self-play and exploration tuning, improving model decision-making efficiency.
- Increased win rate by 22% compared to random and rule-based baselines through iterative refinement.

Sentilyrics: Lyrics Mood Classifier — *NLP, PyTorch, Transformers, BERTs*
- Built an NLP pipeline to classify song lyrics into mood categories for personalized music recommendations.
- Trained and fine-tuned LSTMs, CNNs, and Transformers, achieving 87% accuracy by capturing semantic and emotional cues.
- Implemented preprocessing for 5M+ lyrics, boosting model efficiency and improving F1-score.